

# HBS formation as a second-order phase transition

A Landau functional for SCIANTIX

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September 16, 2026

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## SCIANTIX status

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# SCIANTIX had three formation models for the restructured volume fraction

| option | form                                                            | driven by        |
|--------|-----------------------------------------------------------------|------------------|
| 1      | KJMA, Barani (2020) <sup>1</sup>                                | effective burnup |
| 2      | KJMA + $bu_{inc} = 15$ , Biswas and Aagesen (2025) <sup>2</sup> | effective burnup |
| 3      | KJMA on $\rho_d$ , Veshchunov <sup>3</sup> + Zullo (2026)       | local burnup     |

All three are **transformation kinetics**: a sigmoid fitted in burnup. None of them produces the **mean misorientation**  $\Theta$  of the subgrains, or the **subgrain radius**  $r_n$ .

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<sup>1</sup> Barani et al., *J. Nucl. Mater.* 539 (2020) 152296.

<sup>2</sup> Biswas and Aagesen, *Comput. Mater. Sci.* 258 (2025) 114052, Eq. 45.

<sup>3</sup> Veshchunov and Shestak, *J. Nucl. Mater.* 384 (2009) 12–18.

# The data this model is built on

| paper                                          | technique                               | $\Theta$  | $X$       | $r_n$     |
|------------------------------------------------|-----------------------------------------|-----------|-----------|-----------|
| Zacharie-Aubrun (2022), standard $\text{UO}_2$ | EBSD                                    | 27        | 27        | 14        |
| Zacharie-Aubrun (2022), Cr-doped               | EBSD                                    | 11        | 11        | 8         |
| Onofri (2025)                                  | EBSD, no restructuring ( $\Theta = 0$ ) | 14        | –         | –         |
| Gerczak (2018)                                 | image analysis (Barani 2020), $d_A/2$   | –         | 5         | 3         |
| Noirot (2015)                                  | Xe-depleted area                        | –         | 6         | –         |
| <b>total</b>                                   |                                         | <b>52</b> | <b>49</b> | <b>25</b> |

$$\Theta = [A_{\text{Mis}}(f_1 - f_{10}) + 10f_{10}]/100;$$

$$X = f_{10}, \text{ area fraction above } 10^\circ;$$

$$r_n = \text{ECD50 \%}/2.$$

$1^\circ$  is the lower bin edge of the EBSD data.

# What KJMA is, and what it assumes

Nuclei appear *at random* in the untransformed material and grow *isotropically*; count the transformed volume as if the growing regions could overlap freely, the **extended volume**  $V_e$ , and the overlap correction is statistical:

$$dX = (1 - X) \frac{dV_e}{V} \implies X = 1 - \exp(-K t^n)$$

## What you get

$K$  and  $n$ , and a sigmoid.  $n$  carries the dimensionality of growth and the nucleation regime.

## What it assumes

Randomly placed nuclei, a growth law independent of where you are in the transformation, and a *time dependence*.

# The time dependence in the KJMA model

- 1 **Effective burnup**,  $n = 3.54$ ,  $K = 2.77 \times 10^{-7}$ , fitted on restructured-fraction data.
- 2 Same, with an **incubation burnup**  $bu_{inc} = 15 \text{ MWd/kgHM}$  below which nothing happens. Biswas & Aagesen get it from a balance between the energy stored in dislocations and the energy of forming a subgrain used to shift the origin of an empirical curve rather than to build the curve.
- 3 KJMA not on burnup but on the **dislocation density**  $\rho_d(bu, T)$ , with a threshold  $\rho_{crit} = 6 \times 10^{14} \text{ m}^{-2}$ . The first to make the driver a **microstructural state variable** and the only one that feels temperature.

## What none of them can do

They are calibrated on  $X$  alone. There is no misorientation angle neither subgrain size.

# Why a phase transition could be the right shape for this problem

## The HBS forms *continuously*

Not by nucleating new grains, but by low-angle boundaries absorbing dislocations until their misorientation drifts through the  $10^\circ$  HAGB threshold. In the metallurgical literature this is *continuous* dynamic recrystallization, and it is a well-posed alternative to the nucleation-and-growth picture KJMA describes.<sup>5</sup>

## A continuous transformation has an order parameter

If a single continuous quantity distinguishes the two states, Landau's argument applies: expand the free energy in it, and keep only the terms symmetry allows.<sup>6</sup>

$$F(\eta) = C_0 + C_2\eta^2 + C_4\eta^4$$

Odd powers are forbidden here since the energy of a boundary cannot depend on the *sign* of the misorientation across it.

$C_2$  changing sign produces the transition between a null order parameter and a non-zero one, i.e. the development of misorientation.

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<sup>5</sup>Gourdet & Montheillet, *Acta Mater.* 51 (2003) 2685; Humphreys & Hatherly (2017), chs. 6 and 10.

<sup>6</sup>Landau (1937); Landau & Lifshitz, *Statistical Physics I*, §143.

# Landau against KJMA, side by side

|                           | KJMA (options 1–3)                | Landau (option 4)                                                    |
|---------------------------|-----------------------------------|----------------------------------------------------------------------|
| what is solved            | a <i>kinetic</i> law along a path | the <i>equilibrium</i> of a state                                    |
| driver                    | burnup or $\rho$ threshold        | $\rho(bu)$ through $C_2$                                             |
| free parameters           | $K$ , $n$ fitted on $X$           | $\beta$ , $k$ , $\rho_c$ fitted jointly on $\Theta$ , $X$ ,<br>$r_n$ |
| predicts $X$              | by construction                   | <b>yes</b> , lever rule, no extra parameter                          |
| predicts $\Theta$ , $r_n$ | no                                | <b>yes</b>                                                           |
| reversible?               | no, by construction               | yes, see below                                                       |

## The price of using an equilibrium theory

A drop in  $\rho_{\text{tot}}$  would un-restructure the fuel. Restructuring is irreversible, so the implementation carries a **monotonic lock** on  $\Theta$ . However, this is easy if one considers the dislocation density generated by irradiation, and eventually includes the annealing phenomena within the evolution of the Landau free energy.

## The model

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# HBS as a second-order phase transition

## Order parameter

$$\eta = \frac{\theta}{\theta_{\max}}, \quad \theta_{\max} \equiv \theta_{\text{HAGB}} = 10^\circ$$

the mean misorientation normalised to the LAGB/HAGB boundary.

**External condition:** the local burnup, entering *only* through the dislocation density

$$\rho_{\text{tot}}(bu) = 10^{2.2 \times 10^{-2} bu + 13.8}$$

Nogita & Une (1994)

## Free energy

$$F(bu, \eta) = C_0 + C_2 \eta^2 + C_4 \eta^4$$

Only even powers: the energy cannot depend on the *sign* of a misorientation.

Minimising  $F$  gives  $\Theta$ ; the same wall geometry gives  $r_n$ ; the lever rule gives  $X$ .

## Three adjustable parameters

$\beta$  (wall geometry),  $k$  (sweeping),  $\rho_c$  (strain-field cut-off).

# The geometric wall

A low-angle boundary *is* a wall of dislocations.

**1. Spacing sets the angle**  $\theta = b/d$

so a wall of  $n$  dislocation families carries a line length per unit *wall area*

$$L_A = n\theta/b$$

**2. Wall area per unit volume or low angle grain boundary surface**

$$S/V = 3\sqrt{\rho_{\text{LAGB}}} / \beta$$

$\beta$ : the one geometric parameter (Rest & Hofman 2000).  
 $n = 2$ , Gourdet & Montheillet, not fitted (between 1-3).

**Close the two on each other:**

$$\rho_{\text{LAGB}} = L_A \cdot S/V$$

$$\Rightarrow \sqrt{\rho_{\text{LAGB}}} = \frac{3n\theta}{\beta b}$$

**Everything follows from  $\theta$  alone**

$$\rho_{\text{LAGB}} = \left( \frac{3n\theta}{\beta b} \right)^2, \quad r_n \simeq \frac{\beta^2 b}{6n\theta}$$

# The dislocation balance

Every dislocation is in exactly one of three states, and they must add up to  $\rho_{\text{tot}}$ :

$$\underbrace{\rho_{\text{ord}} = \rho_{\text{LAGB}}^{\text{max}} \eta^2}_{\text{condensed into low-angle walls}} + \underbrace{\rho_{\text{swept}} = (\rho_{\text{tot}} - \rho_{\text{ord}}) \frac{\Delta R}{R}}_{\text{annihilated by moving boundaries}} + \underbrace{\rho_{\text{free}}}_{\text{still a random array}} = \rho_{\text{tot}}$$

Each population carries the line energy of the state it is in:

$$F = \rho_{\text{free}} A_1 G b^2 + \rho_{\text{ord}} A_2 G b^2, \quad A_i = \frac{f(\nu)}{4\pi} \ln \frac{R_i}{b}$$

free ones screened at  $\rho_c$ , wall ones by the wall itself at  $\rho_{\text{tot}}$ ; the swept ones carry nothing.

$$C_0 = \rho_{\text{tot}} A_1 G b^2, \quad C_4 = \rho_{\text{LAGB}}^{\text{max}} \left. \frac{\Delta R}{R} \right|_{\text{max}} A_1 G b^2$$

$$C_2 = \underbrace{\rho_{\text{LAGB}}^{\text{max}} (A_2 - A_1) G b^2}_{\text{walls: the screening gain}} - \underbrace{\rho_{\text{tot}} \left. \frac{\Delta R}{R} \right|_{\text{max}} A_1 G b^2}_{\text{sweep}}$$

**The engine of the transition** is  $A_2 - A_1$ . As burnup rises,  $\rho_{\text{tot}}$  rises, the screening length due to wall formation shrinks,  $A_2$  falls below  $A_1$ , and it becomes cheaper to store dislocations in walls than to leave them random.

## Equilibrium

$$\frac{\partial F}{\partial \eta} = 0 \implies \eta^2 = -\frac{C_2}{2C_4}, \quad \text{clipped at 0}$$

$$\eta = 0 \text{ wherever } C_2 \geq 0.$$

## Condition on the minimisation problem: dislocation balance

The walls cannot hold more dislocations than exist, so  $\rho_{\text{ord}} \leq \rho_{\text{tot}}$ . The equilibrium is the minimum of  $F$  on the **admissible interval**, not the free stationary point:

**Eq. (7b) — admissibility**

$$\eta_{\text{bal}} = \sqrt{\min\left(\frac{\rho_{\text{tot}}}{\rho_{\text{LAGB}}^{\text{max}}}, 1\right)}, \quad \eta_{\text{eq}} = \min\left(\sqrt{-\frac{C_2}{2C_4}}, \eta_{\text{bal}}\right)$$

With the current calibration the bound **never binds** (0 of the 41 standard EBSD points, no burnup at 600 K): it stays in the model as a physical constraint, not as an active branch.

# Calibration

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# The objective and the three data sets

Joint fit of  $\beta, k, \rho_c$  ( $n = 2$  fixed) on all three observables:

$$J = \frac{\langle \Delta \Theta^2 \rangle_w}{\text{var } \Theta} + w_r \frac{\langle \Delta r^2 \rangle_w}{\text{var } r} + w_X \frac{\langle \Delta X^2 \rangle_w}{\text{var } X}$$

dimensionless;

$\langle \cdot \rangle_w$ : average with a weight  $w_i$  per point.

Two questions:

1. **which weights**  $w_r, w_X$ ?
2. **which data set**?

| set | points, weights                                                 |
|-----|-----------------------------------------------------------------|
| A   | Zacharie-Aubrun + Onofri, standard $\text{UO}_2$ ;<br>$w_i = 1$ |
| B   | all four papers; $w_i = 1$                                      |
| C   | as B, $w_i = \text{rank factor} \times \text{relevance}/3$      |

rank factor: A, B  $\rightarrow 1$ ;  $\text{U}^* \rightarrow 0.5$ ;  $\text{U} \rightarrow 0.25$ .

# Where the data come from: quality rank $\times$ relevance

$$w_i = \underbrace{f(\text{Rose rank})}_{\text{how well it was measured}} \times \underbrace{\text{relevance}/3}_{\text{how close to our application}}$$

**1. Rose rank** — M.A. Rose, ANL/CFCT-22/26 (2022) (MSTDB-TP), re-worded for post-irradiation microscopy. Six aspects marked H/M/I: *method, calibration, sample characterisation, irradiation conditions, measurement precision, verifiability*.

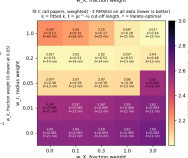
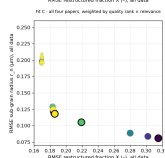
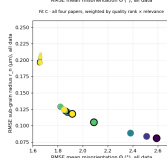
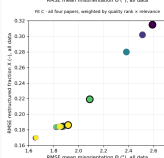
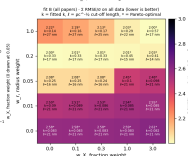
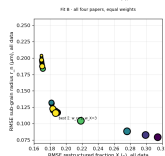
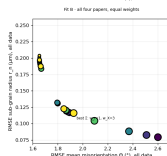
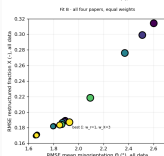
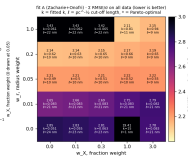
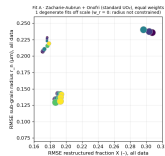
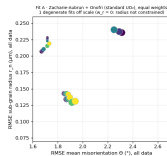
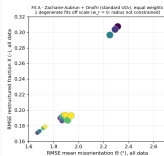
| rank | assigned when                            | $f$  |
|------|------------------------------------------|------|
| A    | all H/M <i>and</i> verifiability H       | 1    |
| B    | $\geq 2$ H/M <i>and</i> verifiab. H/M    | 1    |
| U*   | all but verifiability H/M, unc. unusable | 0.5  |
| U    | uncertainties bar quantitative use       | 0.25 |

**2. Relevance** (1–3).

| sample class               | relevance |
|----------------------------|-----------|
| PWR standard $\text{UO}_2$ | 3         |
| PWR Cr-doped $\text{UO}_2$ | 2         |
| Halden disc, std           | 2         |
| Halden disc, large grain   | 1         |

# Choosing the weights: the calibration front

Calibration front of the Landau model: one fit per data set,  $w, r, \mu$  errors on ALL the data  
marker colour =  $w, X$ , marker size =  $w, r$  (0, 0.05, 0.2, 1); black ring = not dominated by any other fit (all data sets pooled)



# The calibrated parameters

**Set C**,  $w_r = 0.2$ ,  $w_X = 1$

all four papers, each point weighted by  
quality rank  $\times$  relevance

|                  |                                  |
|------------------|----------------------------------|
| $n$ (not fitted) | 2                                |
| $\beta$          | 26.682                           |
| $k$              | 0.5102                           |
| $\rho_c$         | $5.741\text{e}15 \text{ m}^{-2}$ |

**In sample, all four papers**

weighted, with the unweighted beside it

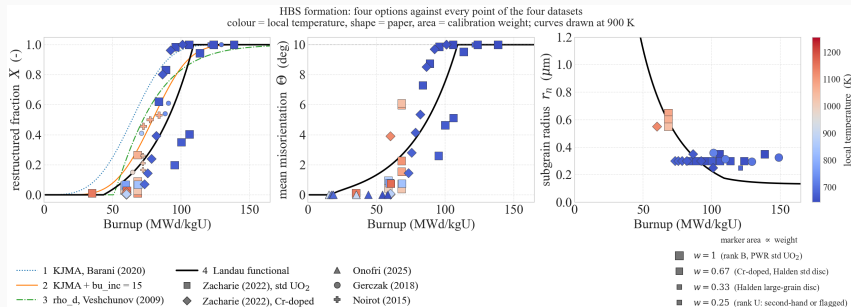
|          | $N$ | RMSE / $R^2$                      | plain                  |
|----------|-----|-----------------------------------|------------------------|
| $\Theta$ | 52  | $1.896^\circ$ / 0.733             | 1.864 / 0.749          |
| $X$      | 49  | 0.194 / 0.723                     | 0.184 / 0.744          |
| $r_n$    | 26  | $0.122 \mu\text{m}$ / <b>0.02</b> | $0.122$ / <b>-0.04</b> |

- $X$  is in the objective.
- The size is still the price:  $R^2 \approx 0$  on all 26 radii — standard  $\text{UO}_2$  alone is +0.47, dragged down by the Cr-doped (-1.41).

## Results

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# Comparison with the other options and with all the data



Restructured fraction  $X$ , all  $N = 49$  (weighted; plain in grey)

| option              | RMSE  | $R^2$ |
|---------------------|-------|-------|
| 1 KJMA              | 0.389 | -0.11 |
| 2 KJMA + $bu_{inc}$ | 0.217 | +0.65 |
| 3 $\rho_d$          | 0.232 | +0.60 |
| 4 Landau            | 0.194 | +0.72 |

plain: 0.380 / 0.204 / 0.227 / 0.184

Option 4 only:  $\Theta$   $1.90^\circ$  / +0.73  
( $N = 52$ ),  $r_n$   $0.122 \mu\text{m}$  / +0.02  
( $N = 26$ ). Fit C, the same weights.

## Against the three models SCIANTIX already had

Burnup [MWd/kgU] at which  $\alpha_r$  first reaches:

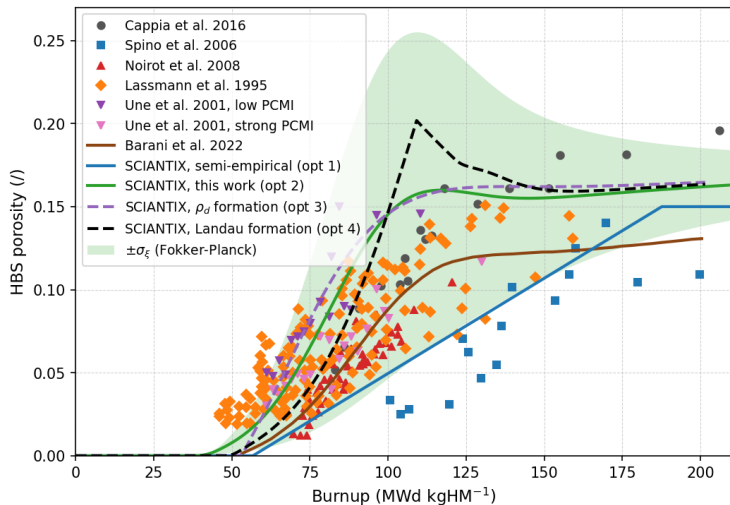
| threshold | 1 KJMA | 2 KJMA+ $bu_{inc}$ | 3 $\rho_d$ | 4 Landau     |
|-----------|--------|--------------------|------------|--------------|
| 1 %       | 19.5   | 34.5               | 51.2       | <b>45.0</b>  |
| 10 %      | 37.8   | 52.8               | 54.8       | <b>58.6</b>  |
| 50 %      | 64.2   | 79.3               | 72.3       | <b>89.2</b>  |
| 90 %      | 90.1   | 105.2              | 112.3      | <b>105.1</b> |
| 99 %      | 109.7  | 124.6              | 161.7      | <b>107.8</b> |

- Option 4 starts **between options 2 and 3** (45 MWd/kgU) but rises more slowly: its 50 % crossing is the **latest of the four**.
- It still **saturates**:  $\Theta$  hits the  $10^\circ$  cap and  $X = 1$  at 108 MWd/kgU, whereas KJMA approaches 1 asymptotically.
- The 1–99 % span is **63 MWd/kgU**, against 90 for option 1 and 110 for option 3.

## Model applied to SCIANTIX

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# Does the new formation model disturb the porosity model?



Largest deviation from option 2 (test\_U02HBS):

**+0.044 absolute at  
95.2 MWd/kgHM  
(+28 %).** Option 4

overshoots to 0.180 at 108 MWd/kgU — its saturation — and dips by **14.5 %** after it. All four options still end within **0.37 %** of the same porosity.

Outlook: from a local  
functional to a phase field

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# What a local functional cannot give -> phase-field models

## Problems of the Landau-like local functional

- no spatial quantities;
- no kinetics: equilibrium is instantaneous.

| family | property                                                                              | application/ref                                                                                                                  |
|--------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| KWC    | two fields ( $\phi, \theta$ ), allows subgrain rotation; singular diffusion equations | Nucleation via subgrain coalescence, Muramatsu (2014)                                                                            |
| MPF    | One order parameter per grain; stable time increments; rotation ignored               | Subgrain growth, Khajezade <i>et al.</i> (2024)                                                                                  |
| HMP    | Removes nonphysical linear gradient term; handles inclination-dependent GB energy     | Coupled Cosserat crystal plasticity, Tandogan <i>et al.</i> (2024); dislocation-driven nucleation, Tandogan <i>et al.</i> (2026) |
| PFC    | Atomic-resolution density field; diffusive timescales                                 | Interface morphology development during solidification, Nit-tala (2025)                                                          |

Nucleation is usually **seeded ad hoc** at a critical density or stress; spontaneous nucleation from lattice-curvature gradients is obtained when coupling KWC with Cosserat crystal plasticity (Ghiglione, 2024), and from stored dislocations at grain boundaries when coupling HMP with it (Tandogan, 2026).

# Muramatsu *et al.* (2014): a KWC phase field for recrystallization

**The functional**<sup>7</sup> — two fields,  $\phi$  and  $\theta$ :

$$\mathcal{F} = \int_V \left[ f(\phi) + \frac{\alpha^2}{2} |\nabla \phi|^2 + s g(\phi) |\nabla \theta| + \frac{\epsilon^2}{2} h(\phi) |\nabla \theta|^2 \right] dV$$

with  $g(\phi) = h(\phi) = \phi^2$ , relaxed by  $\dot{\phi} = -M_\phi \delta \mathcal{F} / \delta \phi$  and  $\dot{\theta} = -M_\theta \delta \mathcal{F} / \delta \theta$ .

$\phi = 1$  inside a subgrain;  $\theta$  is the **lattice orientation**.

The *singular*  $|\nabla \theta|$  term gives a boundary an energy growing with misorientation.

**The local part** (Eq. 7,  $f_m = 0$ ):

$$f(\phi; E_s) = a_{ub}(1 - p(\phi)) E_s + a_{wg} W_b q(\phi)$$

$$q(\phi) = \phi^2(1 - \phi)^2$$

$$p(\phi) = \phi^3(10 - 15\phi + 6\phi^2)$$

- $q$ : the **double well**, original matrix against new grains; its height  $a_{wg} W_b$  sets the boundary energy;
- $p$ : is a sigmoid-like function where  $p'(0) = p'(1) = 0$  therefore it *tilts* the well without moving its two minima;
- $E_s$ : the **stored dislocation energy**, the only driving force, supplied from outside the model.

**How the paper nucleates:** a subgrain group enclosed by high-angle ( $> 10^\circ$ ) boundaries is *selected* as a nucleus candidate and grows after an *incubation period* — the criterion is **imposed**, not produced by the free energy.

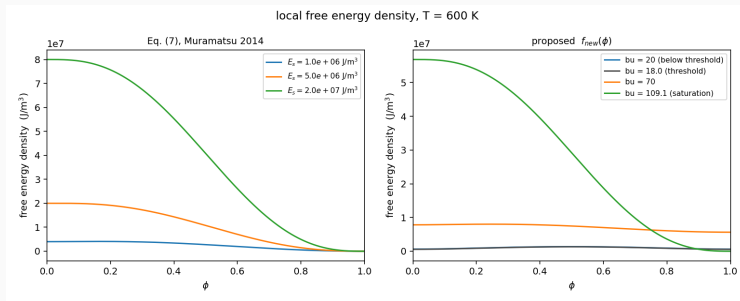
# The prototype: the Landau model supplies the driving force

Keep  $q(\phi)$  and  $p(\phi)$  exactly as above, and replace the externally supplied  $E_s$  by what the Landau model:

$$f_{\text{Mur}}(\phi; E_s) = a_{ub}(1 - p(\phi))E_s + a_{wg}W_b q(\phi)$$

$$f_{\text{new}}(\phi; bu, T) = C_0(1 - p(\phi)\eta) + a_{wg}W_b q(\phi)$$

Both  $C_0$  and  $\eta$  come from the same  $\text{hbs\_state}(bu, T)$  that produces  $\Theta$ ,  $r_n$  and  $X$ .



Left: Eq. (7) at three stored energies. Right:  $f_{\text{new}}$  below, at, and above the threshold.

- Recrystallization is usually modelled with an imposed nucleation (Muramatsu);
- Tandogan *et al.*<sup>8</sup> obtain it spontaneously, from stored dislocations at grain boundaries through a one thermodynamically consistent field theory, Cosserat crystal plasticity + Henry–Mellenthin–Plapp (HMP) orientation phase field (HMP-CCP).

## Order parameters

- $\eta_T \in [0, 1]$ : coarse-grained **crystalline order**, = 1 in the grain bulk, < 1 inside the diffuse GB.
- $\theta$ : lattice orientation, identified with the Cosserat microrotation
- 2D, small strain: dofs  $(\eta_T, \theta, u_1, u_2)$ , FEniCS + Mfront.

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<sup>8</sup>Tandogan, Budnitzki, Sandfeld, *J. Mech. Phys. Solids* 206 (2026) 106325. Presented here up to §3.2.

# The HMP-CCP free energy and the coupling function $\phi(\eta_T)$

$$\psi = f_0 \left[ \alpha V(\eta_T) + \frac{\nu^2}{2} |\nabla \eta_T|^2 + \mu^2 g(\eta_T) |\nabla \theta|^2 \right] + \frac{1}{2} \mathbf{e}^e : \mathbb{E} : \mathbf{e}^e + 2\mu_c (\check{\mathbf{e}}^e)^2 + \underbrace{\phi(\eta_T) \sum_{\alpha} \frac{\lambda}{2} \mu b^2 \rho_{\alpha}}_{\text{stored dislocations}}$$

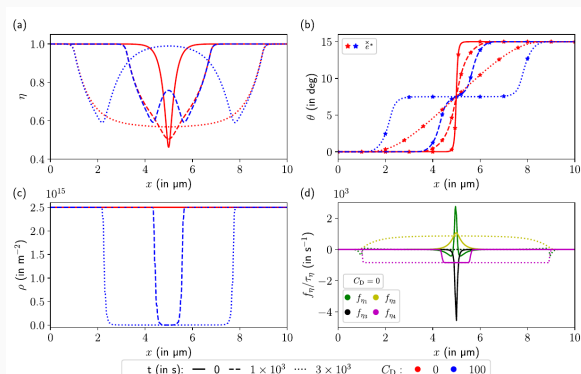
- $V = \frac{1}{2}(1 - \eta_T)^2$ : single well, penalises GBs.
- $g(\eta_T)$  singular at 1, with a  $\ln$  term  $\Rightarrow$  **Read-Shockley** GB energy.
- $\phi_{,\eta} = \frac{c_3}{2} [1 - \tanh(c_1(\eta_T - c_2))]$ : a **step**. The stored energy pushes on  $\eta_T$  only inside the GB ( $\eta_T < c_2$ ); the bulk is “crystalline enough”.
- Recovery (Kocks–Mecking): extra sink  $-\rho C_D A(\eta_T) \dot{\eta}_T$ , active only where  $\dot{\eta}_T > 0$ : **behind a moving GB**.

# The nucleation mechanism: a Cu bicrystal

Periodic bicrystal,  $\Delta\theta = 15^\circ$ , uniform  $\rho_0 = 2.5 \times 10^{15} \text{ m}^{-2}$  switched on at  $t = 0$ .

$C_D = 0$ , **no recovery**: the stored-energy force widens the GB into a non-localised orientation gradient, an *unstable* state.

$C_D > 0$ : recovery at the GB centre: a dislocation-free nucleus bounded by two GBs that move outward.



Tandogan *et al.*, Fig. 4:  $\eta_T$ ,  $\theta$ ,  $\rho$  and the forces on  $\eta_T$ ; red  $C_D = 0$ , blue  $C_D = 100$ .

# Prototype: Tandogan in 1D, driven by the Landau model

**What is solved** — one GB at  $x = L/2$ ,  $L = 10 \mu\text{m}$ , no load ( $u = 0$ , no slip,  $\dot{\epsilon}^e = 0$ ):

$$\tau_\eta \dot{\eta}_T = f_0 \nu^2 \eta_T'' - f_0 [\alpha V' + \mu^2 g' \theta'^2] - \phi_{,\eta} \frac{\lambda}{2} G b^2 \rho$$

$$\hat{\tau}_* g \dot{\theta} = f_0 (\mu^2 g \theta')'$$

$$\dot{\rho} = \textcolor{brown}{S}(\textcolor{brown}{b}u) - \rho C_D A(\eta_T) \langle \dot{\eta}_T \rangle$$

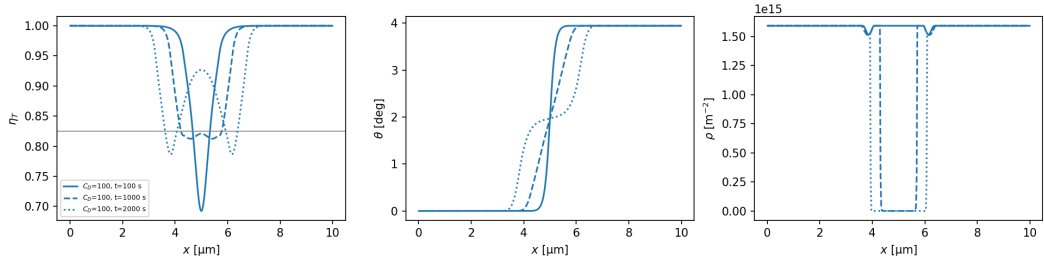
- **Irradiation source** instead of plasticity:  $S = \frac{d\rho_{\text{free}}}{dbu} \dot{b}u$ , uniform, independent of the local  $\rho \Rightarrow$  a recovered region is loaded again.

**From the Landau model:**

- $\Delta\theta \leftarrow \Theta(bu)$ ,  $\rho \leftarrow \rho_{\text{free}}(bu)$  (the random array = Tandogan's SSDs),  $G(T)$ ,  $b_{\text{UO}_2}$ ;
- other parameters as in the paper

# Prototype: what comes out

UO<sub>2</sub>, bu = 80 GWd/tU:  $\Theta = 3.94^\circ$ ,  $\rho_{free} = 1.59\text{e}+15 \text{ m}^{-2}$ ,  $c_3 = 0.52$



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## Backup — the equations

$$(1) \quad \rho_{\text{tot}}(bu) = 10^{2.2 \times 10^{-2} bu + 13.8}$$

$$(2) \quad G(T, P, x, q), \quad \nu(T, P, x, q) \quad \text{NEA/NSC/R(2024)1 p. 124;} \quad f(\nu) = \frac{1-\nu/2}{1-\nu}$$

$$(3) \quad \rho_{\text{LAGB}}^{\text{max}} = \left( \frac{3n\theta_{\text{max}}}{\beta b} \right)^2, \quad (S/V)_{\text{max}} = \frac{9n\theta_{\text{max}}}{\beta^2 b}, \quad \left. \frac{\Delta R}{R} \right|_{\text{max}} = \frac{k \rho_{\text{LAGB}}^{\text{max}}}{\rho_{\text{tot}}}$$

$$(4) \quad A_1 = \frac{f(\nu)}{4\pi} \ln \frac{\rho_{\text{c}}^{-1/2}}{b}, \quad A_2 = \frac{f(\nu)}{4\pi} \ln \frac{\rho_{\text{tot}}^{-1/2}}{b}$$

$$(5,6) \quad F = \rho_{\text{free}} A_1 G b^2 + \rho_{\text{ord}} A_2 G b^2 = C_0 + C_2 \eta^2 + C_4 \eta^4$$

$$(7,7b) \quad \eta_{\text{eq}} = \min(\sqrt{-C_2/2C_4}, \eta_{\text{bal}}), \quad \eta_{\text{bal}} = \sqrt{\min(\rho_{\text{tot}}/\rho_{\text{LAGB}}^{\text{max}}, 1)}$$

$$(8) \quad \Theta = \min(\eta_{\text{eq}} \theta_{\text{max}} \cdot \frac{180}{\pi}, \theta_{\text{HAGB}})$$

$$(9) \quad r_n = \min\left(\frac{1.5}{(S/V)} \left(1 + \frac{\Delta R}{R}\right), R_{\text{grain}}\right)$$

$$(10) \quad X = \text{clip}\left(\frac{\Theta - \theta_u}{\theta_{\text{HAGB}} - \theta_u}, 0, 1 - 10^{-9}\right)$$

# Backup — regimes and parameters

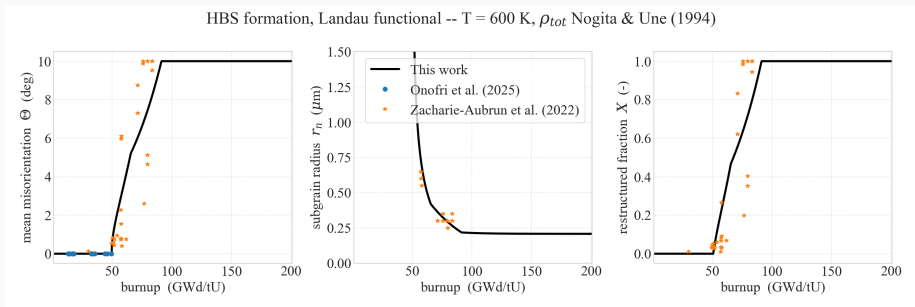
| Regimes ( $T$ -, $P$ -, $x$ -independent) |               |
|-------------------------------------------|---------------|
| condition                                 | $bu$ [GWd/tU] |
| $C_2 = 0$ , $\Theta$ leaves 0             | 18.88         |
| $\Theta = \theta_u$ , $X$ leaves 0        | 42.99         |
| $\Theta = \theta_{\text{HAGB}}$           | 108.07        |

With the set-B parameters the balance bound of Eq. (7b) **never binds**:  $\eta$  is always the stationary point of Eq. (7).

| Fixed constants | $b$                    | 3.889e-10 m              |
|-----------------|------------------------|--------------------------|
|                 | $\theta_{\text{max}}$  | $= \theta_{\text{HAGB}}$ |
|                 | $\theta_{\text{HAGB}}$ | $10^\circ$               |
|                 | $\theta_u$             | $1.00^\circ$             |
|                 | $n$                    | 2                        |

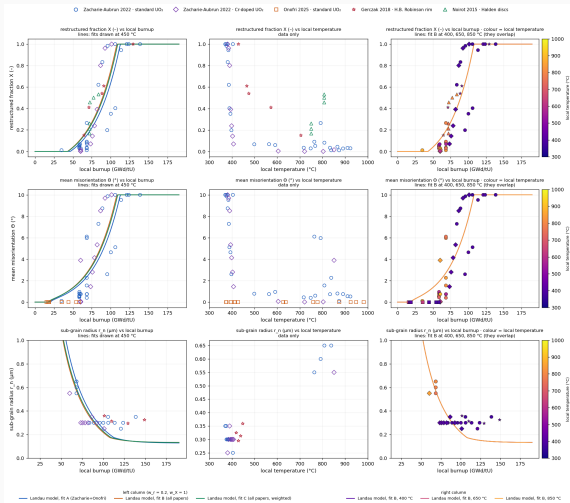
$\theta_u$  is the lower end member of the mixture Eq. (10) inverts, and it is set by the **measurement convention**, not fitted: the EBSD data reports a restructured fraction at  $1^\circ$  ( $f_1$ ) and one at  $10^\circ$  ( $f_{10}$ ), so  $1^\circ$  is the edge below which a boundary is not counted at all. A map whose whole resolved population sits there carries no restructuring — which is  $X = 0$ . It fixes only *where*  $X$  leaves zero:  $\Theta$  and  $r_n$  do not depend on it.

## Backup — the model on its own, $T = 600$ K, $P = 0.05$



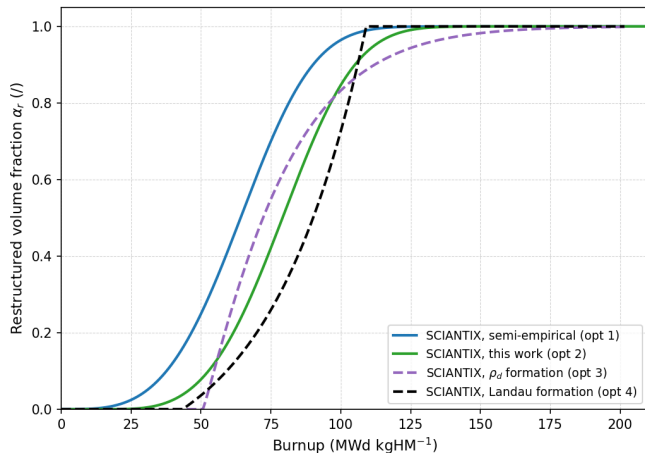
```
python3 hbs_formation_landau.py --plot
```

## Backup — fits A, B, C against the data



```
python3 calibrate.py --study --weight 0.2 --fraction-weight 1
```

## Backup — the four regression cases, as they ship

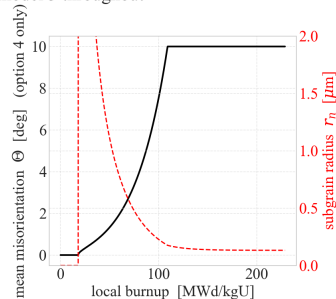
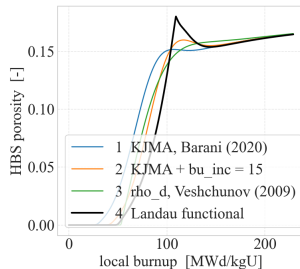
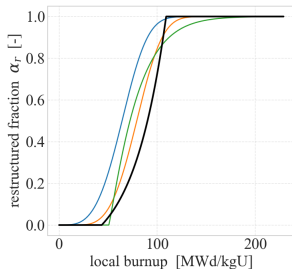


Not the controlled comparison of the previous figure: each case pairs its formation option with the porosity model it was built around (1, 2, 3, 3), so part of the spread is the porosity model.

This is what a user of each regression case actually gets;  
formation\_options.png is what isolates the formation model.

# Backup — the four formation options in a full run

HBS formation options on the same irradiation, porosity model 3 throughout



# Backup — the 1D HMP prototype: implemented equations

Fields  $\eta_{\text{T}}(x, t)$ ,  $\theta(x, t)$ ,  $\rho(x, t)$  on  $x \in [0, L]$ , one GB at  $L/2$ ;  
 $u = 0$ , no slip,  $\theta = -e^*$  (Eq. 27)

$$(32) \quad \tau_{\eta} \dot{\eta}_{\text{T}} = f_0 \nu^2 \eta_{\text{T}}'' - f_0 [\alpha V'(\eta_{\text{T}}) + \mu^2 g'(\eta_{\text{T}}) \theta'^2] - \phi'(\eta_{\text{T}}) \frac{\lambda}{2} G b^2 \rho$$

$$(34) \quad \hat{\tau} g(\eta_{\text{T}}) \dot{\theta} = f_0 (\mu^2 g(\eta_{\text{T}}) \theta')'$$

$$(28) \quad \dot{\rho} = S(bu) - \rho C_D A(\eta_{\text{T}}) \langle \dot{\eta}_{\text{T}} \rangle, \quad \langle \cdot \rangle = \max(\cdot, 0)$$

$$V = \frac{1}{2}(1 - \eta_{\text{T}})^2, \quad A = \frac{7\eta_{\text{T}}^3 - 6\eta_{\text{T}}^4}{1 - \eta_{\text{T}}}$$

$$g = \frac{7\eta_{\text{T}}^3 - 6\eta_{\text{T}}^4}{(1 - \eta_{\text{T}})^3} + c \ln(1 - \eta_{\text{T}}) + g_{\text{off}}, \quad \min g = 0.01$$

$$\phi' = \frac{c_3}{2} [1 - \tanh(c_1(\eta_{\text{T}} - c_2))]$$

$$(42) \quad \eta_{\text{T}}^{\text{eq}} = 1 - \frac{c_3 \frac{\lambda}{2} G b^2 \rho}{f_0 \alpha}$$

$g$ ,  $g'$ ,  $A$  frozen for  $\eta_{\text{T}} > 1 - 10^{-4}$ ;  $\eta'_{\text{T}} = \theta' = 0$  at  $x = 0, L$ .

Coupling to the Landau model, `hbs_state(bu, T)`

$$\Delta\theta = \Theta(bu), \quad \rho \leftarrow \rho_{\text{free}}(bu), \quad G = G(T, \dots)$$

$$c_3 = \frac{(1 - 0.579) f_0 \alpha}{\frac{\lambda}{2} G b^2 \rho_{\text{free}}(bu_{\text{sat}})}$$

so that Eq. (42) gives the paper's Cu value 0.579  
 where  $\Theta = \theta_{\text{HAGB}}$ .

Two protocols for  $\rho$

- *step*:  $S = 0$ ,  $\rho(x, 0) = \rho_{\text{free}}(bu)$  (as in §3.2);
- *irradiation*:  $\rho(x, 0) = 0$ ,  $S = \frac{d\rho_{\text{ref}}}{d bu} \dot{bu}$ ,  
 uniform in  $x$ ,  $\rho_{\text{ref}} \in \{\rho_{\text{free}}, \rho_{\text{tot}}\}$ ,  
 $bu = bu_0 + \dot{bu} t$  (compressed).

Nucleus:  $\rho(L/2) < 0.1 \rho_0$  while  $\min \eta_{\text{T}}(0, L) > c_2$ .

Capillarity (absent in 1D):  $r^* = 2\gamma / (\frac{\lambda}{2} G b^2 \rho)$ .

Interface parameters: Cu, Table 1

$(f_0, \nu, \alpha, \mu, c, \tau_{\eta}, \hat{\tau}, C_D)$ , not calibrated for  $\text{UO}_2$ .

FV, 400 cells, BDF.